MIL-C-43769B
14 October 1988
SUPERSEDING
MIL-M-43769A(GL)
11 July 1984

MILITARY SPECIFICATION

CASE, MAP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers map cases.

1.2 Classification . The map cases shall be of one type and the following sizes as specified (see 6.2):

Size 1 - 22 by 30 inches
Size 2 - 38-1/2 by 52-3/4 inches

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8460

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SPECIFICATIONS

FEDERAL

L-P-375	-	Plastic Film, Flexible, Vinyl Chloride
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-495	-	Tubes, Mailing, and Filing

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-2-538	-	Case, Map

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882 - Tensile Properties of Thin Plastic Sheeting
D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components.

3.2.1 Film. The film used in fabricating all map cases shall be a 0.014 inch thick vinyl conforming to type I, class 1 of L-P-375 with the exception that the film shall meet the requirements for a 0.012 inch thick vinyl film.

3.2.2 Zippered closure. The zipper closure shall be made of a clear vinyl compound conforming to the resin material requirements specified in L-P-375. The zipper shall have a slider and pull of the materials, design and dimensions shown in Drawing 8-2-538.

3.3 Design and construction. The design of the map cases shall be as shown in Drawing 8-2-538. The map cases shall be constructed of two pieces of vinyl film specified in 3.2.1 and a vinyl zipper specified in 3.2.2. The vinyl cases shall be heat sealed on three sides, two long sides and one short side. The open end of the case shall be fitted with a vinyl zipper of appropriate length, heat sealed to the case. One part of the zipper shall be heat sealed to the back of the case and the other part heat sealed to the front part of the case as shown in Drawing 8-2-538. The ends of the zipper shall be heat sealed to prevent the slides from becoming disengaged. Dimensions shall be as shown in Drawing 8-2-538. Tolerances for length and width shall be plus 1 inch, minus 0 inches.

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3.4 Performance.

3.4.1 Sealed seams. All heat sealed seams shall be a minimum of 1/8 inch wide, cleanly trimmed, with no unsealed film beyond permitted sealed area. The breaking strength of the heat sealed seams shall be not less than 16 pounds when tested as specified in 4.4.4.

3.5 Workmanship. The map cases shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I and 4.4.3, and tested for the characteristics specified in 4.4.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of map cases. The sample unit shall be one finished map case. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE I. End item visual defects

Examine	Defect
Material	Not uniformly colorless and clear. Not dull matte finish. Objectionable odor. Any blister or fish eye or unfused resin particles.
Design	Not in accordance with drawing.
Construction	Not in accordance with drawing. Seam less than 1/8 inch wide. Seam not continuous, not uniform. Poor heat seal resulting in inadequate fusion. Corners not rounded.
Workmanship	Any hole, crack, wrinkle, tear, sharp crease or cut. Any spot, stain, or other discoloration which requires a cleaning agent for removal. Not clean, evidence of dirt, grit, oil, or other foreign matter on or in map case. Zipper does not work properly. Any rough or sharp edge.

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4.4.3 End item dimensional examination. The end items shall be examined for compliance with the dimensions annotated with an asterisk on Drawing 8-2-538. Any annotated dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of map cases. The sample unit shall be one map case. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 End item testing. The end items shall be tested for heat sealed seam breaking strength in accordance with method A of ASTM D 882 except that each specimen shall be cut 1 inch by 6 inches with the seam approximately in the middle and parallel to the short sides of the specimen. The lot size shall be expressed in units of map cases. The sample unit shall be one map case and the inspection level shall be S-2. If the average breaking strength for any sample unit fails to meet the requirements specified in 3.4.1, the lot shall be rejected.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of units per container more or less than required

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each map case shall be rolled from the bottom to the slide fastener end. The rolled case shall be placed in a tube conforming to type I, class 1 or 2, style A of PPP-T-495. For size 1 cases, the tube shall have an inside diameter of approximately 2-3/4 inches and a length of approximately 22-1/2 inches. For size 2 cases, the tube shall have an inside diameter of approximately 3-1/2 inches and a length of approximately 39 inches.

5.1.2 Commercial. Each map case shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twenty size 1 cases or ten size 2 cases, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Inside box dimensions for size 1 cases shall be approximately 23 inches in length, 14-1/2 inches in width, and 11-1/2 inches in depth. Inside box dimensions for size 2 cases shall be 39-1/2 inches in length, 18 1/4 inches in width, and 7-1/2 inches in depth. Each box shall be closed, waterproofed, and reinforced with strapping or tapebanding in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Twenty size 1 cases or ten size 2 cases, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, double wall (DW), type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The fiberboard box liner shall conform to type CF, class domestic, variety DW, grade 275 of PPP-P-636. Inside box dimensions for size 1 cases shall be approximately 23 inches in length, 14-1/2 inches in depth. Inside box dimensions for size 2 cases shall be approximately 39-1/2 inches in length, 18-1/4 inches in width, and 7-1/2 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with method III, as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Map cases, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), map cases, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding O or P. Pallet pattern shall be number 90 for size 1 cases, and number 89 for size 2 cases in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. Map cases are used to protect maps used in the field.

6.2 Order data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size of map case desired.
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load (see 5.2.1).
f. When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Case
Map

6.5 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU

Review activity:

DLA - CT

User activity:

Navy - MC

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43769B(GL)	Case, Map

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK (P), 1 Apr 84

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